L04: Entities, Resources, Events, Activities, and State

Simulation Without a Black Box

Outline

Learning Objectives

By the end of this lecture you will be able to:

- Define entities and their attributes with concrete examples.
- Characterize resources by capacity, discipline, and preemption rules.
- Distinguish events (instantaneous) from activities (duration-spanning).
- Identify the minimal state variable set needed to resume a simulation.
- Use standard notation W, W_q, L, L_q, ρ correctly.
- Describe the eight components of a complete conceptual model document.

Entities and Their Attributes

Entity

A distinguishable object that flows through the system. Its life cycle is traced from creation to destruction.

System	Entity	Typical attributes
Hospital ED	Patient	Arrival time, acuity level, insurance type
Manufacturing line	Job	Arrival time, processing steps, priority
Call center	Call	Arrival time, call type, patience threshold
Port terminal	Container	Weight, destination, arrival ship ID
Inventory system	Order	Quantity demanded, arrival time

Attributes are recorded *per entity*. They are not system-level counters.

Resources: Capacity, Discipline, Preemption

Capacity

Maximum simultaneous users. A physician pool of capacity 3 can serve 3 patients at once; the 4th waits.

Service discipline

Rule for selecting the next entity from the queue.

- FCFS (First Come, First Served) — default
- Priority (acuity level; shortest job first)
- LCFS, round-robin, random

Preemption

Can a higher-priority entity interrupt an in-progress service?

- **Preemptive resume**: interrupted service resumes from where it stopped.
- **Preemptive restart**: service restarts from the beginning.
- **Non-preemptive**: current service always completes.

In SimPy: use `resource.request(priority=k)` with `PriorityResource` or `PreemptiveResource`.

Events vs. Activities

Event (instantaneous)

Occurs at a single point in simulated time; changes state immediately.

- Arrival of a patient
- Service completion
- Machine breakdown
- Inventory order placed

Activity (duration-spanning)

Has a beginning event and an ending event with time in between. During an activity, the relevant resource is busy.

- Service (service-begin $\rightarrow$ service-end)
- Travel (departure $\rightarrow$ arrival at next station)
- Repair (breakdown $\rightarrow$ restoration)

DES implementation

Both are needed. Activities are represented as a *pair* of events separated by a sampled duration. In SimPy, `yield env.timeout(duration)` spans an activity.

State Variables: Minimal Sufficient Set

State variable

A quantity whose value at time t is necessary and sufficient to reproduce all future system behavior given future inputs.

Single-server queue example:

- $N_q(t)$: number of customers in queue (not counting server)
- $B(t) \in \{0, 1\}$: server busy (1) or idle (0)
- $T_{next}(t)$: scheduled end time of current service (if $B = 1$)

What you do NOT need to store:

- Full history of past service times (unless computing output statistics)
- Individual customer IDs (unless tracking attributes)

A bloated state slows the simulation and obscures verification.

Performance Measures: Standard Notation

Symbol	Meaning	How computed from simulation
W	Mean time in *system* per customer	Average of (departure − arrival)
W_q	Mean time in *queue* per customer	Average of (service-begin − arrival)
L	Mean number in *system*	Time-average of $N(t)$
L_q	Mean number in *queue*	Time-average of $N_q(t)$
ρ	Server utilization	Fraction of time server is busy
λ	Effective arrival rate	Customers served / simulation time

These six quantities are linked by Little's Law: $L = \lambda W$ and $L_q = \lambda W_q$.

Eight Components of a Conceptual Model Document (§3.5)

1. **Problem statement** — decision and objectives.
2. **System boundary** — what is in/out; endogenous/exogenous.
3. **Entity table** — entity types and their attributes.
4. **Resource table** — capacity, discipline, preemption.
5. **Event list** — all event types and their triggers.
6. **State variable list** — minimal sufficient set.
7. **Performance measures** — output variables and their formulas.
8. **Assumptions and simplifications** — what is omitted and why.

Why formalize this?

The document is the *contract* between the analyst and the stakeholder. Verification checks that the code matches the document; validation checks that the document matches reality.

Activity: Entity/Resource Table for the Primary-Care Clinic

Complete the following table for the primary-care clinic from L03.

Element	**Type**	**Attributes / Capacity**	**Discipline**
Patient	Entity	Arrival time, acuity, appt. type	—
Check-in clerk	Resource	Capacity: ______	FCFS
Exam room	Resource	Capacity: ______	____________
Physician	Resource	Capacity: ______	Priority (acuity)
Arrival	Event	Schedules next arrival; joins queue	—
Service end	Event	________________________	—

Add two more entities or resources you would include in a medium-scope model.

Key Takeaways

- Entities have attributes; resources have capacity, discipline, and preemption rules.
- Events are instantaneous; activities span time and are bounded by a pair of events.
- The state variable set must be minimal — only what is needed to resume simulation.
- Standard notation: $W, W_q, L, L_q, \rho, \lambda$ — learn these symbols; they are used throughout.
- A complete conceptual model document has eight components. The code should be a faithful translation of the document, not the other way around.

Next lecture: The simulation clock and event-driven time advance (L05).